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Enhancement of the material point method using B-spline basis functions

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The material point method (MPM) enhanced with B-spline basis functions, referred to as B-spline MPM (BSMPM), is developed and demonstrated using representative quasi-static and dynamic example problems. Smooth B-spline basis functions could significantly reduce the cell-crossing error as known for the original MPM. A Gauss quadrature scheme is designed and shown to be able to diminish the quadrature error in the BSMPM analysis of large-deformation problems for the improved accuracy and convergence, especially with the quadratic B-splines. Moreover, the increase in the order of the B-spline basis function is also found to be an effective way to reduce the quadrature error and to improve accuracy and convergence. For plate impact examples, it is demonstrated that the BSMPM outperforms the generalized interpolation material point (GIMP) and convected particle domain interpolation (CPDI) methods in term of the accuracy of representing stress waves. Thus, the BSMPM could become a promising alternative to the MPM, GIMP, and CPDI in solving certain types of transient problems.

KEYWORDS
B-spline basis functions, Gauss quadrature, large deformation, material point method, mesh refinement, transient problems

1 | INTRODUCTION

The material point method (MPM) is a continuum-based particle method, which was motivated by the need for better simulating impact and penetration problems[1,2] and has evolved into a robust spatial discretization scheme for multiphase (fluid-gas–solid) interactions involving failure evolution as shown in a recent comprehensive review.[3] In the MPM, the continuum bodies are represented by discrete Lagrangian particles (or points), which carry the material properties such as mass, strain, stress, internal state variables, etc., and are tracked throughout the deformation process of the bodies. As compared with other meshfree particle methods, 1 striking difference of the MPM is the use of a background Eulerian grid for solving the governing equations. Owing to the combined usage of the particles and the grid, the MPM possesses the advantages of both Lagrangian and Eulerian methods. Since its first journal paper,[1] the

MPM has been further developed and applied to many challenging problems in simulation-based engineering science (SBES) by many research teams over the past 2 decades, such as high-speed impact and penetration, landslide, saturated porous media, fluid–structure interaction, and thermomechanical damage (eg,[4-12]).

There are 2 particular issues encountered in the original MPM. First, piece-wise linear grid basis functions are used to map the properties between the particles and the grid nodes and to calculate the internal and external forces at the grid nodes. Due to the discontinuous gradients of linear basis functions at the grid cell boundary, the so-called "cell-crossing instability" would occur when the particles cross a cell boundary, resulting in the nonphysical jump in the internal forces and velocity gradient and thus reducing the solution accuracy.[13] Second, the MPM suffers from the quadrature error in the calculation of quantities at grid nodes.[14] It arises from the fact that the moving material particles are chosen as the integration points in the standard MPM, instead of Gauss points as used in the finite element method (FEM).

To reduce the cell-crossing error, Bardenhagen and Kober[13] developed the generalized interpolation material point (GIMP) method, in which the weighting functions are defined using the particle characteristic function to replace the linear basis functions for the information mapping between the particles and the grid nodes and for the nodal force calculations. As compared with the C^0 linear basis functions, the weighting functions are C^1 continuous, and their gradients have no discontinuity at the cell boundary. Numerical examples have shown that the GIMP is able to reduce the cell-crossing error and to improve the solution accuracy, which therefore suggests that the use of high-order continuous grid basis functions could be an effective way for reducing the cell-crossing error at additional computational expense.

The pioneering concept of isogeometric analysis as proposed by Hughes et al[15] has gained significant success in computational mechanics (eg,[16-20]), making B-spline basis functions a highly promising and appealing candidate for the grid basis functions in the MPM. Steffen et al[14] studied the spatial convergence of the MPM algorithms using quadratic and cubic B-spline basis functions, and demonstrated that the adoption of B-spline basis functions could decrease the internal force errors caused by cell crossing. Tielen[21] recently performed the numerical tests of a vibrating bar and a column under self-weight using the quadratic basis functions and confirmed the same finding. However, his numerical solutions also exhibited significant oscillations in the particle stresses for large-deformation cases, which was ascribed to the material particle-based integration as mentioned previously. Hence, Tielen[21] recommended the incorporation of Gauss quadrature into the MPM with the aid of the function reconstruction technique, but the focus was limited to the quadratic basis functions with the very coarse grids up to 64 cell elements for 1-dimensional cases.

The purpose of this work is to investigate and compare the accuracy and convergence of the B-spline MPM (BSMPM) schemes using quadratic, cubic, and quartic B-spline basis functions, respectively, in the context of a wide range of grid resolutions. In addition, the performance of the BSMPM in simulating impact problems with strong stress jumps and its comparison with that of the GIMP, which has not been made in the previous studies,[14,21] will also be evaluated. The remainder of this paper begins with an overview of the MPM and B-spline basis functions. The formulation of the BSMPM and corresponding Gauss quadrature are then described, followed by the BSMPM simulations of representative problems, through which the convergence and improvement of the BSMPM are demonstrated and discussed. Finally, concluding remarks and suggestions for future work are presented.

2 | B-SPLINE BASIS FUNCTIONS AND BSMPM

2.1 | MPM

To be self-contained, we first provide a brief description for the "modified update stress last" (MUSL) version of the explicit MPM algorithm,[2] which is employed in the BSMPM to be presented later. As mentioned before, the MPM discretizes the continuum domain into a finite collection of material particles, each of which has a fixed mass and carries the material properties as well as its own constitutive relation. The particles are placed within an Eulerian background grid used for solving the governing equations and updating the variables at the particles. For simplicity, the grid is usually defined as a regular Cartesian mesh with equal spacing. Suppose that all particle properties (eg, position, velocity, density, stress, etc.) after k time cycles are known. The MPM computation in the $k+1$-th cycle proceeds as follows, with the subscripts i and p denoting the quantities associated with the grid nodes and particles, respectively. The first step is to map the mass and velocity of particles to the corresponding grid nodes, namely

$$m_i^k = \sum_p m_p S_i\left(\boldsymbol{x}_p^k\right) \tag{1}$$

$$\boldsymbol{v}_i^k = \frac{\sum_p m_p \boldsymbol{v}_p^k S_i\left(\boldsymbol{x}_p^k\right)}{m_i^k} \tag{2}$$

where m_i and $\boldsymbol{v}_i$ are the mass and velocity vector at grid node i, respectively, m_p and $\boldsymbol{v}_p$ are the mass and velocity vector at material particle p, respectively, $S_i(x_p)$ is the basis function at grid node i with respect to the particle position vector of $\boldsymbol{x}_p$, the superscript k refers to the k-th time cycle, and the subscripts i and p run over all the grid nodes and material particles, respectively. The internal force vector at grid nodes, $\boldsymbol{f}_i^{\text{int}}$, is calculated by

$$\left(\boldsymbol{f}_i^{\text{int}}\right)^k = -\sum_p \boldsymbol{G}_i\left(\boldsymbol{x}_p^k\right)\cdot\boldsymbol{\sigma}_p^k V_p^k \tag{3}$$

in which $\boldsymbol{G}_i(\boldsymbol{x}_p)$ is the gradient of basis function $S_i(\boldsymbol{x}_p)$, $\boldsymbol{\sigma}_p$ is the stress tensor evaluated at particle p, and $V_p = m_p/\rho_p$ is the volume of particle p with ρ_p being the particle density in its current configuration.

The essential and natural boundary conditions are then applied, and the nodal acceleration vector $\boldsymbol{a}_i$ is obtained by

$$\boldsymbol{a}_i^k = \frac{\left(\boldsymbol{f}_i^{\text{int}}\right)^k + \left(\boldsymbol{f}_i^{\text{ext}}\right)^k}{m_i^k} \tag{4}$$

where $\boldsymbol{f}_i^{\text{ext}}$ is the external force vector applied at grid node i. Hence, the nodal velocity is updated using the explicit Euler forward time integration, ie,

$$\boldsymbol{v}_i^{k+1} = \boldsymbol{v}_i^k + \boldsymbol{a}_i^k \Delta t \tag{5}$$

with Δt being the time step that satisfies the stability criterion.

Next, the updated particle velocity and position vectors are found by mapping the nodal data back to the corresponding particles,

$$\boldsymbol{v}_p^{k+1} = \boldsymbol{v}_p^k + \Delta t \sum_i S_i\left(x_p^k\right)\boldsymbol{a}_i^k \tag{6}$$

$$\boldsymbol{x}_p^{k+1} = \boldsymbol{x}_p^k + \Delta t \sum_i S_i\left(x_p^k\right)\boldsymbol{v}_i^{k+1} \tag{7}$$

After that, the updated particle velocity $\boldsymbol{v}_p^{k+1}$ is projected to the nodes for the nodal velocity $\overline{\boldsymbol{v}}_i^{k+1}$ that is used to calculate the strain increment at particles in conjunction with the application of essential boundary conditions, namely

$$\overline{\boldsymbol{v}}_i^{k+1} = \frac{\sum_p m_p \boldsymbol{v}_p^{k+1} S_i\left(x_p^k\right)}{m_i^k} \tag{8}$$

$$\Delta\boldsymbol{\varepsilon}_p = \frac{\Delta t}{2}\sum_i \left\{\overline{\boldsymbol{v}}_i^{k+1}\boldsymbol{G}_i\left(\boldsymbol{x}_p^k\right) + \left[\overline{\boldsymbol{v}}_i^{k+1}\boldsymbol{G}_i\left(\boldsymbol{x}_p^k\right)\right]^T\right\} \tag{9}$$

in which $\Delta\boldsymbol{\varepsilon}_p$ is the strain increment tensor at particle p. Note that all the mapping between the nodal and particle quantities are performed using the basis functions evaluated at the particle position $\boldsymbol{x}_p^k$.

Finally, for given strain increments, the corresponding stress increments are computed with the constitutive relation, and the particle stress tensor is updated as

$$\boldsymbol{\sigma}_p^{k+1} = \boldsymbol{\sigma}_p^k + \Delta\boldsymbol{\sigma}_p^{k+1} \tag{10}$$

with $\Delta\boldsymbol{\sigma}_p$ representing the stress increment tensor at particle p.

2.2 | B-spline basis functions

The description of B-spline basis functions starts with the univariate case and then extends to the multivariate cases.[15,17,22,23] The univariate B-spline basis functions are constructed based on a *knot vector*, which is defined as an ordered sequence of nondecreasing knot values $\Xi=\{\xi_1, \xi_2, \ldots, \xi_{n+p}, \xi_{n+p+1}\}$, where the relation of $\xi_1 \leq \xi_2 \leq \ldots \leq \xi_{n+p} \leq \xi_{n+p1}$holds, ξ_i is the i-th knot with $i = 1, 2, \ldots, n + p + 1$, n is the number of basis functions, and integer p is the polynomial order. The nonzero intervals $[\xi_i, \xi_{i+1})$ are named as knot span. The B-spline basis functions of order $p = 0, 1, 2, 3, 4$, etc. are referred to constant, linear, quadratic, cubic, quartic, etc. B-spline basis functions, respectively. A knot vector is uniform when the knots are equally spaced, and otherwise, it is non-uniform. As shown in the definition of knot vector, more than 1 knot can be at the same location. A knot has multiplicity k if it is repeated k times. Particularly, a knot vector is said to be open if its first and last knots appear $p + 1$ times, namely, their multiplicity $k = p + 1$. Open knot vectors are commonly used in the computer-aided design and analysis, and the BSMPM in this work is also developed using the B-spline basis functions based on them.

Given a knot vector, the associated univariate B-spline basis functions of order p, $N_{i,p}$, can be defined by the well-known Cox-de Boor recursion formula.[24] The zeroth order basis functions are piecewise constants ($p = 0$) as follows:

$$N_{i,0} = \begin{cases} 1 & \text{if } \xi_i \leq \xi < \xi_{i+1} \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

For $p \geq 1$, the basis functions are defined by the recursion scheme,

$$N_{i,p}(\xi) = \frac{\xi - \xi_i}{\xi_{i+p} - \xi_i} N_{i,p-1}(\xi) + \frac{\xi_{i+p+1} - \xi}{\xi_{i+p+1} - \xi_{i+1}} N_{i+1,p-1}(\xi) \tag{12}$$

where the assumption of $0/0 = 0$ is made. Equations (11) and (12) give a set of n basis functions with the following properties:

1. They form a partition of unity, $\sum_{i=1}^{n} N_{i,p} = 1$.
2. They are non-negative, $N_{i,p}(\xi) \geq 0$, for all ξ.
3. $N_{i,p}(\xi)$ is supported in the interval $[\xi_i, \xi_{i+p+1}]$.

The derivative of $N_{i,p}(\xi)$ can be also computed by a recursive formula,[23,24] ie,

$$\frac{dN_{i,p}(\xi)}{d\xi} = \frac{p}{\xi_{i+p} - \xi_i} N_{i,p-1}(\xi) - \frac{p}{\xi_{i+p+1} - \xi_{i+1}} N_{i+1,p-1}(\xi) \tag{13}$$

The B-spline basis functions constructed from an open knot vector are interpolatory at the end knots of ξ_1 and ξ_{n+p+1}, but generally not interpolatory at the interior knots, as illustrated in Figure 1A for the quadratic B-spline basis functions based on an open uniform knot vector Ξ={0, 0, 0, 1, 2, 3, 3, 3}. The B-spline basis functions of order p by a uniform knot vector are C^{p-1}-continuous. At a knot of multiplicity k, the basis functions become C^{p-k}-continuous. When a knot has multiplicity p, the basis functions are interpolatory at this knot. For example, Figure 1B illustrates

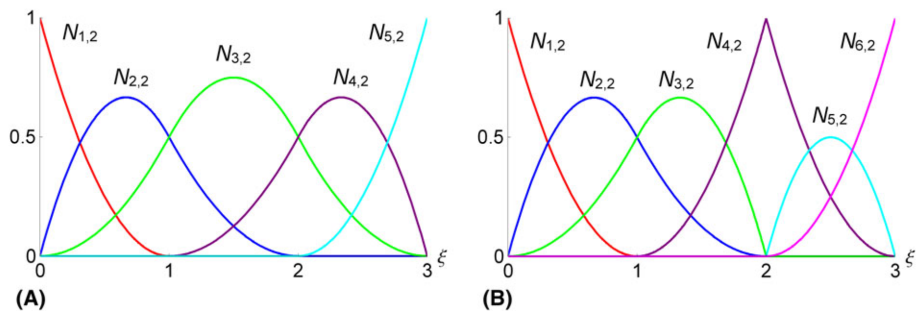

FIGURE 1 Quadratic B-spline basis functions for (A) open and uniform knot vector Ξ={0, 0, 0, 1, 2, 3, 3, 3}, and (B) open, non-uniform knot vector Ξ={0, 0, 0, 1, 2, 2, 3, 3, 3}

the quadratic basis functions constructed from an open, nonuniform knot vector $\Xi = \{0, 0, 0, 1, 2, 2, 3, 3, 3\}$. In addition to the end knots, the basis functions are also interpolatory at $\xi = 2$, due to the repeated knots of ξ_5 and ξ_6 at this location.

The multivariate B-spline basis functions can be built from the univariate ones by means of tensor product structure. In the bivariate case, given knot vectors $\Xi = \{\xi_1, \xi_2, \ldots, \xi_{n+p}, \xi_{n+p+1}\}$ and $\boldsymbol{H} = \{\eta_1, \eta_2, \ldots, \eta_{m+q}, \eta_{m+q+1}\}$ along the parametric directions of ξ and η, respectively, where p and q are the polynomial order, and n and m are the numbers of basis functions along ξ and η, respectively, the tensor-product bivariate B-spline basis functions are expressed as[15]

$$N_{(i,j),(p,q)}(\xi, \eta) = N_{i,p}(\xi) N_{j,q}(\eta) \tag{14}$$

where $i = 1, 2, \ldots, n$ and $j = 1, 2, \ldots, m$. The trivariate B-spline basis functions can be derived in a fashion similar to that in the bivariate case. In addition to knot vectors Ξ and $\boldsymbol{H}$, an additional knot vector $\boldsymbol{L}$ corresponding to the third parametric coordinate ζ is given as $\boldsymbol{L} = \{\zeta_1, \zeta_2, \ldots, \zeta_{l+r}, \zeta_{l+r+1}\}$. The trivariate B-spline basis functions in 3 dimensions are given as

$$N_{(i,j,k),(p,q,r)}(\xi, \eta, \zeta) = N_{i,p}(\xi) N_{j,q}(\eta) N_{k,r}(\zeta) \tag{15}$$

where $k = 1, 2, \ldots, l$ and r is the polynomial order corresponding to the parametric coordinate ζ.

2.3 | Formulation of the BSMPM

It is known that the cell-crossing error due to the use of linear Lagrangian basis functions is the main source for the degraded accuracy of the original MPM. Attempts have been made to improve the MPM using higher-order Lagrangian basis functions.[25] However, it was shown that the use of higher-order Lagrangian basis functions is not only unable to effectively reduce the crossing error but also results in the numerical instability due to nonphysical negative nodal mass.[21,25] Instead, B-spline basis functions have the properties of non-negativity, partition of unity, and higher order of continuity. Therefore, it is natural to develop the BSMPM approach by simply replacing Lagrangian basis functions with B-spline basis functions, without any other changes being made to the explicit MUSL MPM algorithm as described in Subsection 2.1.

For the purpose of simplicity and clear demonstration, the BSMPM concept is presented in the 2-dimensional framework, from which the 1-dimensional and 3-dimensional cases could be easily derived. As schematically illustrated in Figure 2A, 2 open knot vectors of Ξ and $\boldsymbol{H}$ are defined with the ordered knot sequences of $\xi_1 = \cdots = \xi_{p+1} < \xi_{p+2} < \cdots < \xi_{n+1} = \cdots = \xi_{n+p+1}$ and $\eta_1 = \cdots = \eta_{q+1} < \eta_{q+2} < \cdots < \eta_{m+1} = \cdots = \eta_{m+q+1}$, respectively, and used to build a parametric grid Ω, in which the particles (solid blue circles) are embedded. A tensor product grid $\widehat{\Omega}$ is further constructed with the total of $n \times m$ nodes (see Figure 2B), which are in 1-to-1 correspondence with the B-spline basis

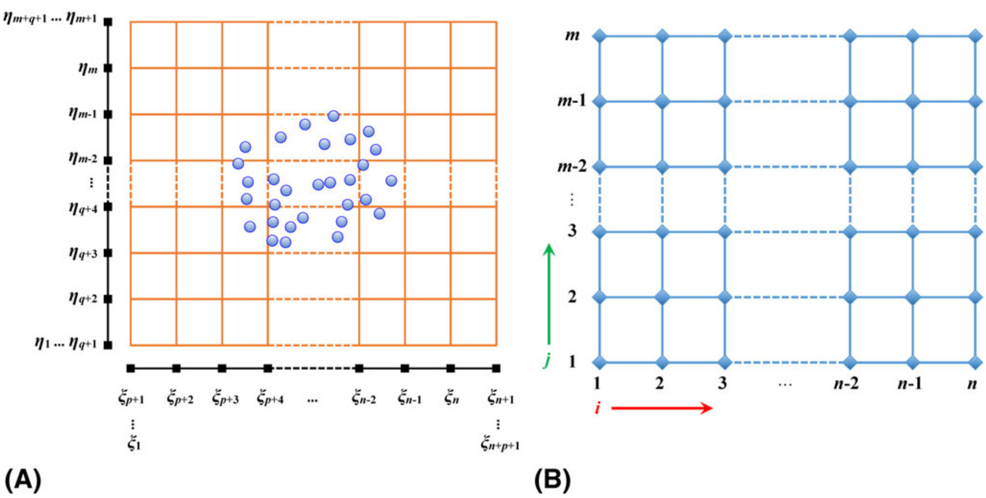

FIGURE 2 Illustrations for (A) the parametric grid of Ω and (B) the tensor product grid of $\widehat{\Omega}$ in 2 dimensions as defined based on the open knot vectors of Ξ and $\boldsymbol{H}$. The solid blue circles and solid black squares in the parametric grid denote the discrete particles and the knots, respectively. The tensor product grid nodes are represented by diamond-shaped points

functions as defined in Equation (14). In other words, the basis function $N_{(i,j),(p,q)}(\xi, \eta)$ corresponds to a unique node with the index of (i, j) in $\widehat{\Omega}$. In the BSMPM, the B-spline basis functions for the particles are determined via the topological relation between the particles and the knots in parametric grid Ω and then used to project the particle properties onto the tensor product grid nodes in $\widehat{\Omega}$. Equations (1)–(3) are rewritten as follows:

$$m_{(i,j)}^{k} = \sum_{p} m_p N_{(i,j),(p,q)}\left(\boldsymbol{x}_p^k\right) \tag{16}$$

$$\boldsymbol{v}_{(i,j)}^{k} = \frac{\sum\limits_{p} m_p \boldsymbol{v}_p^k N_{(i,j),(p,q)}\left(\boldsymbol{x}_p^k\right)}{m_{(i,j)}^{k}} \tag{17}$$

$$\left(\boldsymbol{f}_{(i,j)}^{\text{int}}\right)^{k} = -\sum_{p} \nabla N_{(i,j),(p,q)}\left(\boldsymbol{x}_p^k\right) \cdot \boldsymbol{\sigma}_p^k V_p^k \tag{18}$$

where the subscript (i, j) denotes the quantities at node (i, j) in $\widehat{\Omega}$. The momentum equation is solved on the tensor product grid nodes by the same procedure as that in the MPM, and then the solutions at the tensor product grid nodes are mapped back to the particles to update the particle properties using the B-spline basis functions:

$$\boldsymbol{v}_p^{k+1} = \boldsymbol{v}_p^k + \Delta t \sum_{i} N_{(i,j),(p,q)}\left(\boldsymbol{x}_p^k\right) \boldsymbol{a}_{(ij)}^k \tag{19}$$

$$\boldsymbol{x}_p^{k+1} = \boldsymbol{x}_p^k + \Delta t \sum_{i} N_{(i,j),(p,q)}\left(\boldsymbol{x}_p^k\right) \boldsymbol{v}_{(ij)}^k \tag{20}$$

Similar to the MPM, the updated particle velocity is projected to the tensor product grid nodes for calculating the strain increments at particles:

$$\overline{\boldsymbol{v}}_{(i,j)}^{k+1} = \frac{\sum\limits_{p} m_p \boldsymbol{v}_p^{k+1} N_{(i,j),(p,q)}\left(\boldsymbol{x}_p^k\right)}{m_{(i,j)}^{k}} \tag{21}$$

$$\Delta\boldsymbol{\varepsilon}_p = \frac{\Delta t}{2} \sum_{i} \left\{ \overline{\boldsymbol{v}}_{(i,j)}^{k+1} \nabla N_{(i,j),(p,q)}\left(\boldsymbol{x}_p^k\right) + \left[\overline{\boldsymbol{v}}_{(i,j)}^{k+1} \nabla N_{(i,j),(p,q)}\left(\boldsymbol{x}_p^k\right)\right]^T \right\} \tag{22}$$

As can be seen, the physical space is discretized by the parametric grid, while the tensor product grid is defined for indexing the basis functions and facilitating the computer coding process. Consequently, the spatial size of knot span in the parametric grid should be used to determine the critical time step for the BSMPM based on the Courant-Friedrichs-Lewy (CFL) condition.

The parametric grid can be refined by inserting new knots between 2 distinct knots already in the knot vector,[15] which is an analogue to the classical h-refinement in the finite element analysis. An example of knot insertion is given in Figure 3. Four quadratic B-spline basis functions constructed from an open uniform knot vector $\Xi = \{0, 0, 0, 1, 2, 2, 2\}$ are illustrated in Figure 3A. A new knot vector of $\{0, 0, 0, 0.5, 1, 1.5, 2, 2, 2\}$ is formed by inserting 2 new knots at 0.5 and 1.5 into Ξ, from which 6 quadratic B-spline basis functions are built, as shown in Figure 3B. The spatial convergence of the BSMPM to be discussed later corresponds to the refinement of the parametric grid by knot insertion.

3 | GAUSS QUADRATURE IN THE BSMPM

In addition to the cell-crossing error, the quality of the MPM also suffers from the quadrature error in the internal force calculation.[14] A method based on Gauss quadrature rule can be used in the BSMPM to calculate internal forces for the reduced quadrature error.[21] In general, the expression of internal forces $\boldsymbol{f}_i^{\text{int}}$ takes the form of

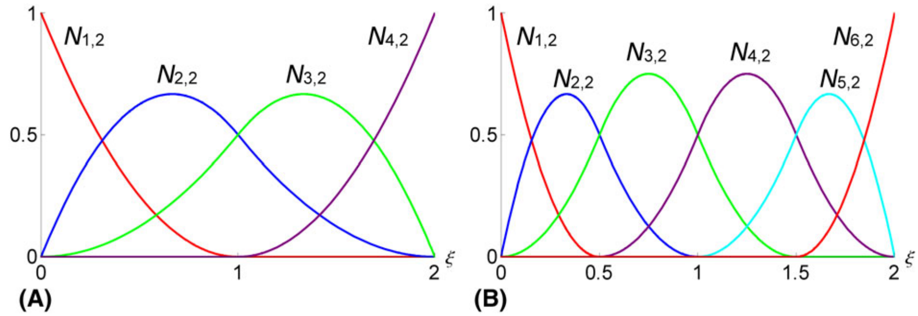

FIGURE 3 Refinement of the parametric grid by knot insertion: (A) quartic basis functions for the original knot vector of {0, 0, 0, 1, 2, 2, 2}, and (B) quartic basis functions for the new knot vector of {0, 0, 0, 0.5, 1, 1.5, 2, 2, 2} formed by inserting 2 additional knots into the original knot insertion

$$\boldsymbol{f}_i^{\text{int}} = -\int_{\Omega_t} \rho \boldsymbol{\sigma}^s \nabla S_i(\boldsymbol{x}) d\Omega \approx -\sum_{g=1}^{n_g} \omega_g \boldsymbol{\sigma}(\boldsymbol{x}_g) \nabla S_i(\boldsymbol{x}_g) \tag{23}$$

where Ω_t denotes the particle domain, $\boldsymbol{\sigma}^s$ is the specific stress tensor, n_g is the number of integration points, and ω_g and $\boldsymbol{x}_g$ are the weight and the position vector of integration points, respectively. Equation (23) is reduced to Equation (3) when using the particle as the integration point and the particle volume as the weight.

To apply the Gauss integration, the particle domain is simply approximated as a line segment, rectangle, and cuboid for 1-dimensional, 2-dimensional, and 3-dimensional cases, respectively. The particle is positioned at the domain center and the density is uniformly distributed over the domain. Depending on the problem dimension, the particle domain is equally divided into 2, 4, or 8 subdomains, respectively, as shown in Figure 4A. The internal forces are calculated by utilizing the Gauss quadrature rule in each subdomain. Figure 4B illustrates the Gauss integration scheme for the 1-dimensional case, in which x_p and l_p are the coordinate of the particle and the length of the particle domain, respectively. The length of l_p is tracked at every time step via the deformation gradient of the particle, which is usually computed for updating the particle stress. The particle domain is divided into 2 equal subdomains of $[x_p\text{-}l_p/2, x_p]$ and $[x_p, x_p+l_p/2]$. Based on Equation (23), the internal forces are calculated by

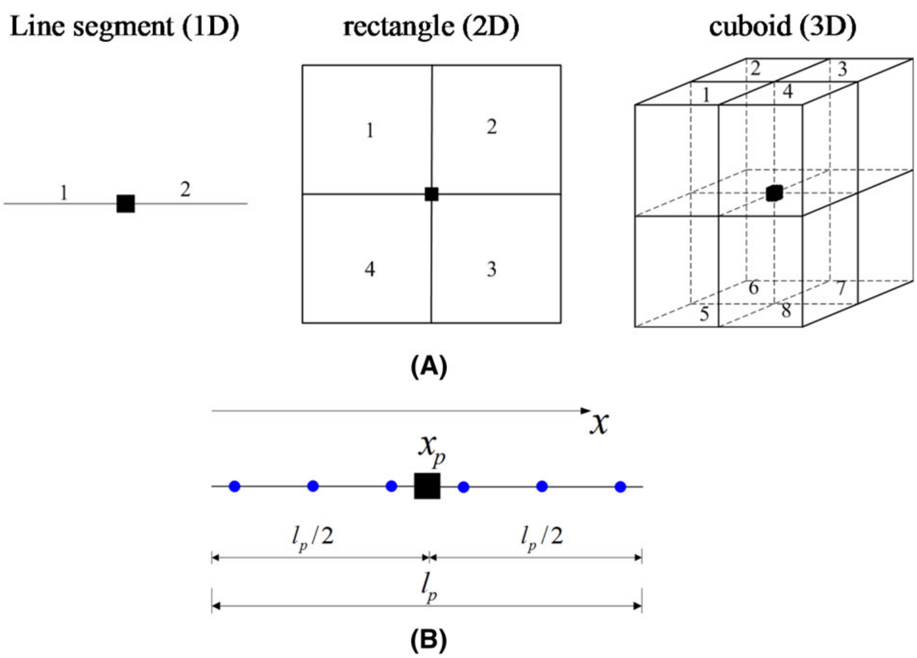

FIGURE 4 Schematic illustrations for (A) the division of the 1-dimensional, 2-dimensional, and 3-dimensional particle domains and (B) the Gauss point integration in the particle subdomain. The solid black squares and solid blue circles represent the particle and the Gauss points, respectively

$$\left(\boldsymbol{f}_i^{\text{int}}\right)^k = -\frac{l_p}{4}\sum_{c=1}^{n_c}\omega_c\boldsymbol{\sigma}\left(\frac{l_p}{4}x_c + x_p - \frac{l_p}{4}\right)\nabla S_i\left(\frac{l_p}{4}x_c + x_p - \frac{l_p}{4}\right) \\ -\frac{l_p}{4}\sum_{c=1}^{n_c}\omega_c\boldsymbol{\sigma}\left(\frac{l_p}{4}x_c + x_p + \frac{l_p}{4}\right)\nabla S_i\left(\frac{l_p}{4}x_c + x_p + \frac{l_p}{4}\right) \tag{24}$$

where n_c is the number of Gauss points, and ω_c and $\mathbf{x}_c$ are the weight and the position of Gauss points (denoted by blue solid circles in Figure 4B), respectively. Note that the stresses are only evaluated at the particles in the BSMPM. The stresses at the integration points could be obtained by the cubic spline interpolation of particle stresses.[24] For multidimensional cases, the integral for internal forces in Equation (23) is also numerically evaluated by the Gauss point quadrature technique with respect to each dimension, and the moving least squares (MLS) method could be an alternative to the cubic spline interpolation scheme for finding the stresses at Gauss points. The details of the cubic spline interpolation and the MLS can be found elsewhere (eg,[26-28]) so that they are not given here. Note that the particle domains in the 2 and 3 dimensions could be more accurately tracked and approximated as quadrilaterals and hexahedra using the convected particle domain interpolation (CPDI) method,[29,30] allowing for the further improved calculation of internal forces.

4 | NUMERICAL EXAMPLES

Several representative examples are adopted to evaluate and demonstrate the validity and convergence of the proposed BSMPM. The first 2 examples are a vibrating bar with the fixed ends and a column under self-weight, which were used in[21] as benchmark problems. In addition to the dynamic response, the quasi-static case of the column due to self-weight is also studied by the BSMPM. Two plate impact examples are then considered to compare the capabilities of the BSMPM, GIMP, and CPDI methods in simulating the large-deformation impact response and to examine the applicability of the MPM contact algorithm to the BSMPM.

4.1 | Vibration of a bar with the fixed ends

As shown in Figure 5, an elastic bar of length L is fixed at its 2 ends. An initial longitudinal velocity of $v(x) = v_0 \sin(\pi x/L)$ is applied to the whole bar to induce its vibration. The material properties and parameters are as follows: Length $L = 25$ m, Young's modulus $E = 100$ Pa, density $\rho = 1$ kg/m^3, and $v_0 = 0.1$ m/s. According to,[21] the bar will have small deformations.

To construct the B-spline basis functions, open uniform knot vectors, $\Xi = \left\{\overbrace{0, \ldots, 0}^{p+1}, \Delta x, 2\Delta x, \ldots, L-2\Delta x, L-\Delta x, \overbrace{L, \ldots, L}^{p+1}\right\}$, are defined, with Δx denoting the length of knot span. The material particles are initialized with respect to the knot spans. Let nps denote the initial number of particles per knot span. The bar is discretized by dividing each knot span into nps subregions and placing 1 material particle at the centroid of each sub-region. In addition to the BSMPM simulations, the problem is also solved using the original MPM, in which the grid cell size is also equal to Δx and the material particles are initialized in the same way as that for the BSMPM. We choose $\Delta x = 0.125$ m and $nps = 4$ for all simulations, and the quadratic, cubic, and quartic B-splines basis functions are used in the BSMPM calculations. The time steps for the BSMPM and the MPM are both 1×10^{-4} seconds. The corresponding Courant number is 8.0×10^{-3}, indicating that the adopted time step is sufficiently small such that the error due to temporal discretization is negligible as compared with that caused by the spatial discretization.

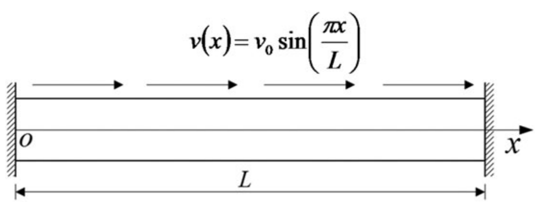

FIGURE 5 The example problem of bar vibration

The accuracy of the BSMPM and the MPM is assessed via the L^2-error norm Err_2 defined as

$$\mathrm{Err}_2 = \frac{\sqrt{\sum_{p=1}^{n_p}\left[\overline{R}(x_p,\ t) - R^{\mathrm{sim}}(x_p,\ t)\right]^2}}{\sqrt{\sum_{p=1}^{n_p}\left[\overline{R}(x_p,\ t)\right]^2}} \tag{25}$$

where n_p is the total number of material particles, and $\overline{R}$ and R^{sim} are the reference and simulation results at the particles, and via the L^∞-error norm Err_∞ defined as

$$\mathrm{Err}_\infty = \frac{\max\limits_{1 \le p \le n_p}\left|\overline{R}(x_p,\ t) - R^{\mathrm{sim}}(x_p,\ t)\right|}{\max\limits_{1 \le p \le n_p}\left|\overline{R}(x_p,\ t)\right|} \tag{26}$$

The analytical solutions are available for this small-deformation problem and taken as a reference. Based on the wave propagation theory, the analytical displacement $\overline{u}(x,\ t)$ and stress $\overline{\sigma}(x,\ t)$ in the bar at time t are given as[31]

$$\overline{u}(x,\ t) = \frac{v_0 L}{\pi c}\sin\left(\frac{\pi c}{L}t\right)\sin\left(\frac{\pi x}{L}\right) \tag{27}$$

$$\overline{\sigma}(x,\ t) = E\frac{v_0}{c}\sin\left(\frac{\pi c}{L}t\right)\cos\left(\frac{\pi x}{L}\right) \tag{28}$$

where $c = \sqrt{E/\rho}$ is the elastic wave speed in the bar.

Figure 6 compares the distributions of particle displacement and stress along the bar axis at time $t = 0.5$ seconds as obtained by the MPM and the BSMPM with the analytical solutions. It is shown in the figure that the displacements obtained by the MPM and the BSMPM all match the analytical solution well. The stress comparisons reveal 2 jumps in the MPM stresses at $x =$ ~4.5 and ~20 m, with noticeable deviations from the analytical solutions. As indicated by

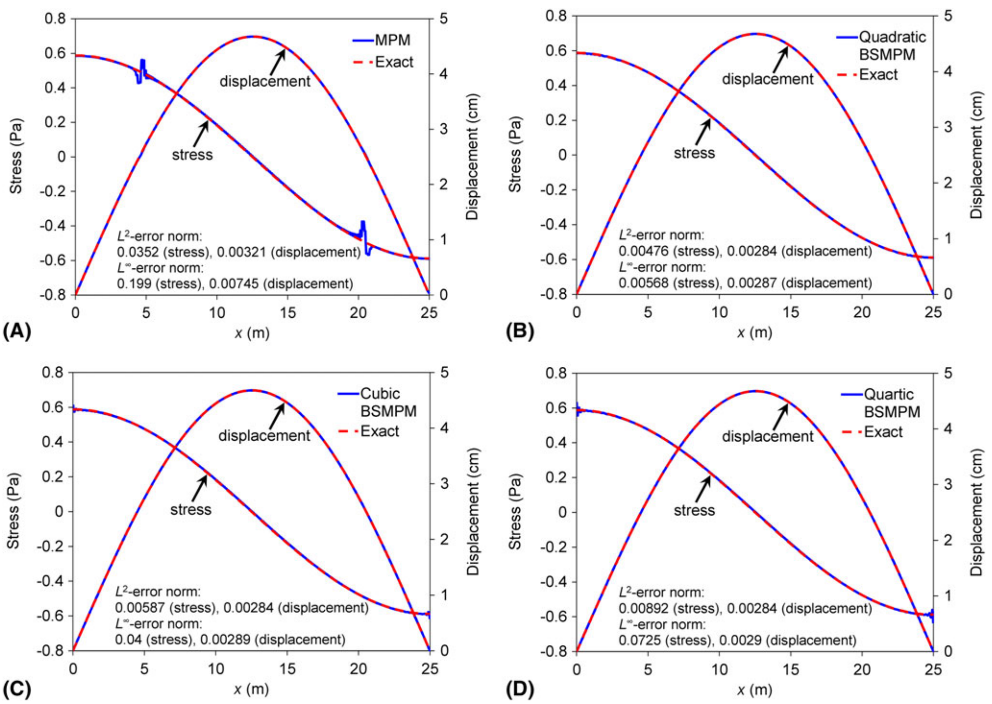

FIGURE 6 Comparisons of the distributions of particle displacement and stress along the bar axis at 0.5 s calculated by the MPM and the BSMPM with the analytical solutions: (A) MPM, (B) quadratic BSMPM, (C) cubic BSMPM, and (D) quartic BSMPM

the L^2-error norms, the BSMPM calculations yield highly improved stress distributions in a good agreement with the analytical results, showing the superior accuracy of the BSMPM over the MPM.

Meanwhile, the oscillations of particle stresses at the boundaries are clearly observed for the cubic and quartic BSMPM, especially for the higher-order quartic case, as shown by the L^∞-error norms given in the figure. The use of consistent mass matrix to project the particle velocities has been suggested to reduce the stress oscillations at the boundaries[21] at the expense of a considerably increased computational cost devoted to solving a linear system of equations. Alternatively, we propose to simply place more particles in the boundary spans. The problem is then solved using the quartic BSMPM scheme with the same parameters as those for Figure 6D, except $nps = 16$ in the boundary spans of $[0,\Delta x)$ and $[L\text{-}\Delta x, L)$, and the corresponding stress profiles at $t = 0.5$ seconds are plotted in Figure 7. It can be seen from the figure that by assigning more particles in the boundary spans, the stress oscillations at the boundaries are considerably reduced as compared with the results in Figure 6D, with the L^∞-error norm decreasing from 0.0725 to 0.0159.

We have further monitored the potential energy $U(t)$ and the kinetic energy $K(t)$ of the vibrating bar within the first 30 seconds. The total mechanical energy $E(t)$ is the sum of $U(t)$ and $K(t)$. Figure 8 illustrates the time histories of total mechanical energy ratio of $E(t)/E(0)$ and the kinetic energy ratio of $K(t)/K(0)$ calculated by the MPM and the BSMPM, as well as the corresponding analytical solutions.[32] As shown in Figure 8B-D, all the BSMPM predictions are in reasonable agreement with the analytical results. The BSMPM has the good energy conservation property with the total energy difference being less than 0.01% from the analytical results. By contrast, Figure 8A shows a large discrepancy in the kinetic energy between the MPM and the analytical results after the first cycle of vibration. The maximum difference in total energy between the MPM and the analytical formulas is ~0.12%.

4.2 | A column under self-weight

As shown in Figure 9, an elastic column of height $H = 1$ m is fixed at the bottom and starts to deform at time $t = 0$ due to the self-weight of ρg. The column has Young's modulus $E = 1.0 \times 10^5$ Pa and density $\rho = 1.0 \times 10^3$ kg/m^3, the same as those for the large-deformation case in.[21] The space discretization of the column follows the same procedure as that in the previous vibrating bar example. We choose $\Delta x = 0.005$ m and discretize the column with 4 particles per knot span or MPM cell. Following the method demonstrated in the vibrating bar example, an increased number of 16 particles is initialized in the boundary span of $[0,\Delta x)$ for the BSMPM calculations to reduce the stress oscillations at the fixed boundary. Furthermore, a time step of 2.0×10^{-5} seconds is chosen for all simulations, and then a Courant number of 0.04 is obtained. Because no analytical solutions are available for this large-deformation problem, the FEM results obtained with 4000 elements are adopted as a reference. Our numerical tests show that the changes in the stresses and displacements are, respectively, less than 0.07% and 0.06% as the element number is increased from 4000 to 8000.

Figure 10 presents the comparisons of the particle displacements and stresses in the column at 0.5 seconds as computed by the BSMPM and MPM with the reference finite element results. As can be seen, the MPM fails to describe the column responses with poor predictions of the displacement and stress distributions in the column. In comparison, the BSMPM computations, for any order of the B-spline basis functions, provide the displacement results agreeing well

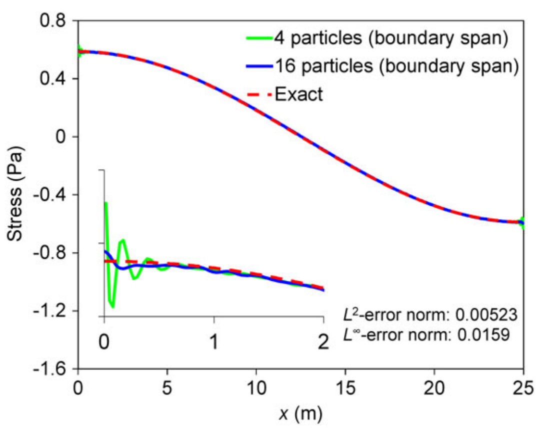

FIGURE 7 Comparisons of the stress profiles along the bar axis at 0.5 s obtained by the analytical solutions with those by the quartic BSMPM calculations using $nps = 4$ and 16 in the boundary spans. The inset is the enlarged view of the stress profiles in the boundary range of $x = 0\text{-}2$ m. The results with $nps = 4$ in the boundary spans are the stress distribution in Figure 6D

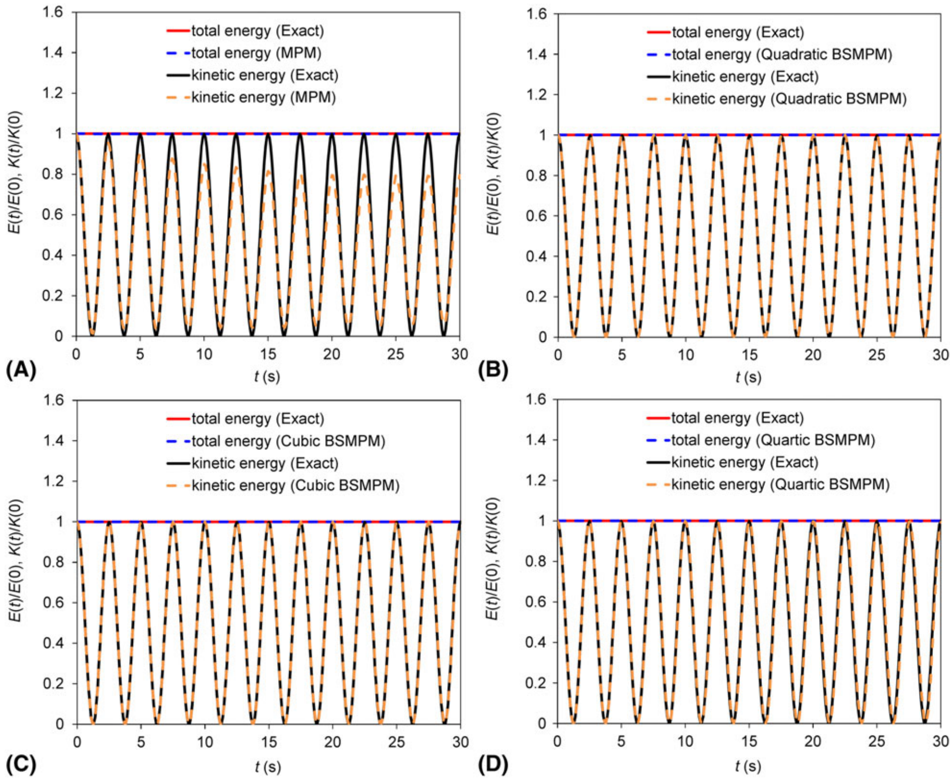

FIGURE 8 Comparisons of the total mechanical energy ratio of $E(t)/E(0)$ and the kinetic energy ratio of $K(t)/K(0)$ for the vibrating bar as obtained with the MPM, BSMPM, and analytical method: (A) MPM, (B) quadratic BSMPM, (C) cubic BSMPM, and (D) quartic BSMPM

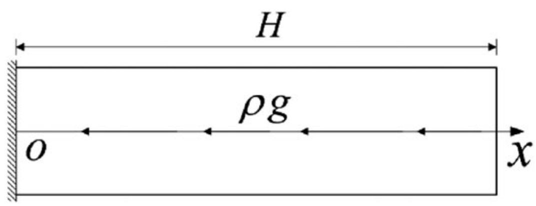

FIGURE 9 The problem of an elastic column under self-weight

with the FEM solutions. The use of B-spline basis functions also leads to the improvement of stress results, the extent to which depends on the basis function order. When using the quadratic B-spline basis functions, there are significant oscillations in the simulated stresses over the whole column. For the cubic and quartic BSMPM, the stress oscillations are greatly reduced, and the simulated stresses favorably agree with the FEM results. Particularly, the stress oscillations are hardly seen in the quartic basis case.

Due to their higher order of continuity, B-spline basis functions overcome the problem of cell-crossing error. The observed stress oscillations in the quadratic BSMPM are thus believed to most likely result from the quadrature error that is not severe in the small-deformation cases. To clarify this issue, we have further simulated the problem with the BSMPM involving the 3 Gauss point quadrature for the internal force calculations, in which the cubic spline interpolation technique is utilized for the stress reconstruction. It is shown in Figure 11 that the stress oscillations are significantly reduced as a result of using the Gauss point quadrature scheme, particularly obvious for the quadratic BSMPM, and that the newly computed stresses agree well with the FEM results for all the BSMPM calculations. This confirms that it is the quadrature error that leads to the notable stress oscillations over the column as shown in Figure 10A for the quadratic BSMPM. According to the results in Figures 10 and 11, we can find that the reduction of the quadrature error for improving the BSMPM accuracy could be achieved by using the presented Gauss quadrature approach and/or increasing the order of B-spline basis functions.

The quasi-static response for the column due to self-weight has also been studied here. By following the loading procedure in,[13,32] the self-weight load is linearly increased from zero to the specified value in 10 seconds, which is 100 times

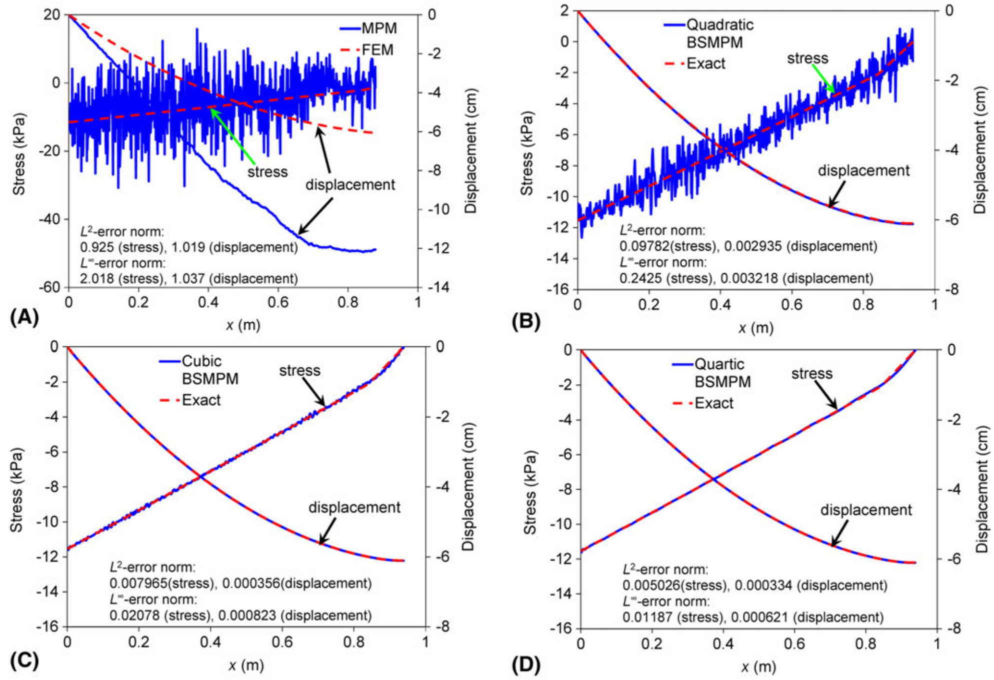

FIGURE 10 Comparisons of the distributions of particle displacement and stress along the column axis at 0.5 s calculated by the MPM and the BSMPM with the reference FEM results: (A) MPM, (B) quadratic BSMPM, (C) cubic BSMPM, and (D) quartic BSMPM. The errors are calculated using Equations (25) and (26)

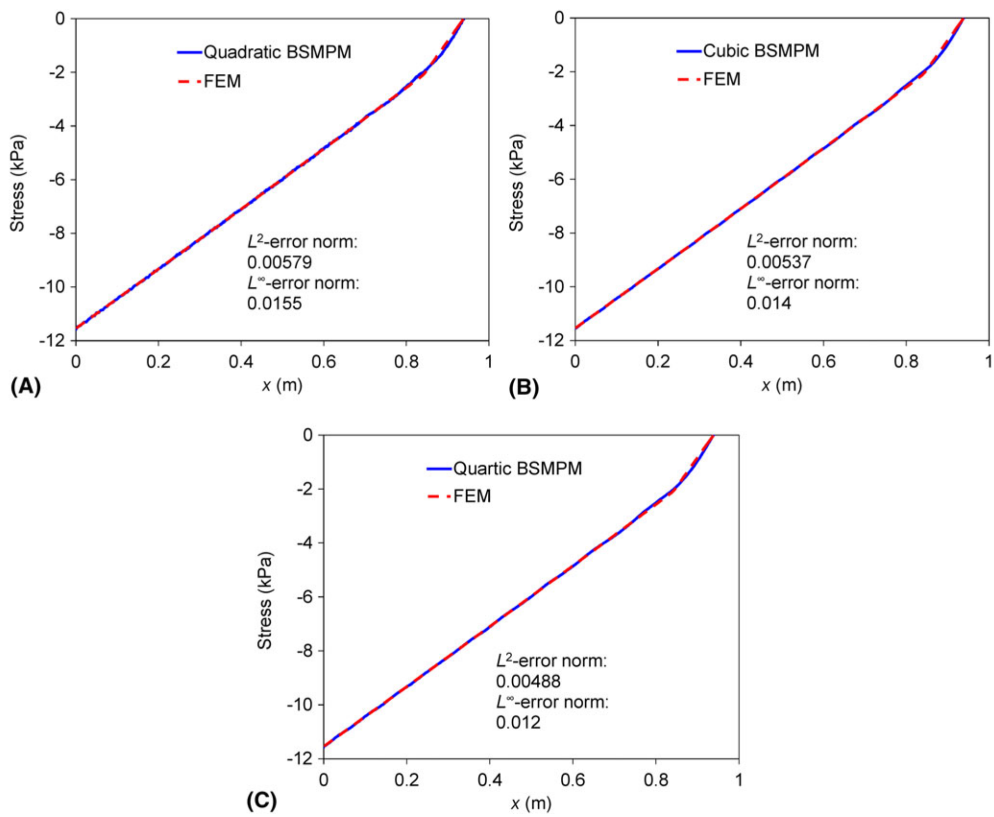

FIGURE 11 Comparisons of the stress distributions over the column at 0.5 s calculated by the (A) quadratic, (B) cubic, and (C) quartic BSMPM involving Gauss quadrature with the FEM results. The errors are calculated using Equations (25) and (26)

the elastic wave propagation time in the column so that it is long enough to obtain a quasi-static state. The gravitational acceleration is purposely set at 49.05 m/s^2 to generate the maximum engineering strain of ~50%. The other parameters remain the same as those in the dynamic case. The distributions of particle stresses along the column at 5 and 10 seconds with the MPM and the BSMPM without Gauss quadrature are plotted in Figure 12. As illustrated in Figure 12A, the particle stresses obtained with the MPM are quite noisy. The BSMPM solutions as shown in Figure 12B-D are in good agreement with the analytical results,[32] except for the slight oscillations in the quadratic BSMPM case.

4.3 | Plate impact problems

Figure 13A schematically illustrates the impact of an elastic plate against a rigid wall at an initial velocity v_0 along its thickness direction. We study the stress wave propagations in the plate thickness direction using the BSMPM and GIMP. Because the planar dimensions of the plate are much larger than its thickness, the problem could be simplified as a uni-axial strain case along the thickness direction, as illustrated in Figure 13B. The plate has a thickness of $H = 0.06$ m, and its Young's modulus E, Poisson's ratio ν, density ρ, and initial velocity are 1.0×10^5 Pa, 0.3, 1000 kg/m^3, and 0.6 m/s, respectively.

To construct the B-spline basis functions, open uniform knot vectors are defined with the length of knot span $\Delta x = 1 \times 10^{-4}$ m. The material particles are initialized with $nps = 4$, except $nps = 16$ in the leftmost knot span. In the GIMP computations, the grid is composed of uniform cells with a size equal toΔx, and the discretization is performed by setting 4 particles per cell. Two GIMP strategies, namely, uniform/unchanged GIMP (uGIMP) and contiguous particle GIMP (cpGIMP), have been proposed.[33] The former strategy assumes the unchanged size of particles, while the latter allows the particle size to evolve according to the deformation gradient at the particle. Because the plate will experience large deformations, the cpGIMP scheme is adopted to investigate the stress waves in the plate. The time step for all the BSMPM and GIMP simulations is 1.0×10^{-6} seconds, and the impact begins at $t = 0$ seconds.

To smooth the oscillations at the shock front, an artificial viscosity q, which contains linear and quadratic viscosity terms, is added to the particle stresses in the internal force calculations, and defined as[34-36]

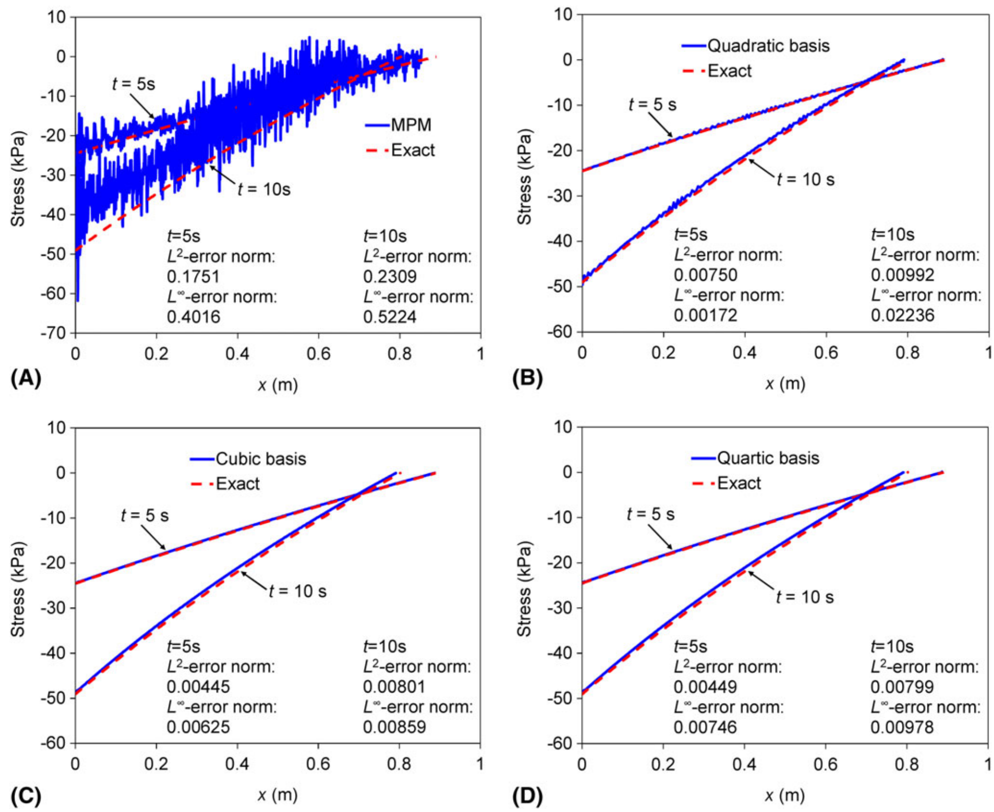

FIGURE 12 Comparisons of the particle stresses along the column as obtained with the MPM, BSMPM and analytical method for the quasi-static problem of a column under self-weight: (A) MPM, (B) quadratic BSMPM, (C) cubic BSMPM, and (D) quartic BSMPM

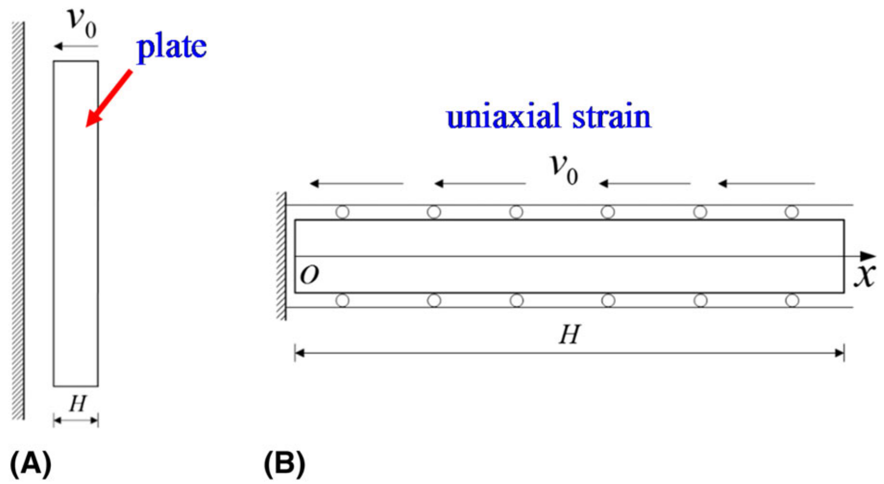

FIGURE 13 Schematic illustration of the plate impact against a rigid wall: (A) side view of the problem, and (B) simplification as a uniaxial strain case along the plate thickness direction

$$\begin{cases} q = \rho L_c\left(-c_0 a\dot{\varepsilon}_v + c_1 L_c \dot{\varepsilon}_v^2\right) & \dot{\varepsilon}_v<0 \\ q = 0 & \dot{\varepsilon}_v \geq 0 \end{cases} \tag{29}$$

in which L_c is the characteristic length, c_0 and c_1 are coefficients for the linear and the quadratic viscosities, respectively, $a = \sqrt{\frac{E(1-\nu)}{\rho(1+\nu)(1-2\nu)}}$ is the local sound speed, and $\dot{\varepsilon}_v$ is the volumetric strain rate. In this uniaxial strain problem, characteristic length L_c is chosen to be the particle volume, and $\dot{\varepsilon}_v$ is equal to the strain rate along the thickness direction.

The distributions of the stress along the plate thickness at 4 different time instants as computed with the GIMP, CPDI, and the cubic BSMPM involving the 3 Gauss point integration are compared in Figure 14, where the artificial viscosity is applied with $c_0 = 3$ and $c_1 = 4$. For a better comparison, the FEM-calculated stress distributions based on 4000 elements are also included in the figure as a reference. It can be seen that the computed wave front positions and stress magnitudes with the cubic BSMPM are in good consistence with those given by the FEM, except the stress magnitudes around the wave fronts due to the dissipation by the artificial viscosity. Although good GIMP and CPDI predictions of the wave front positions are present, notable oscillations can be found in the compressive stresses computed with the GIMP and CPDI.

One might attribute the oscillating compressive stresses in the GIMP and CPDI to the insufficient artificial viscosity. To clarify this issue, further GIMP and CPDI simulations have been performed using the increased coefficients of c_0 and c_1, and the corresponding stress distributions at 7.8 ms are shown in Figure 15. As the artificial viscosity is increased, the oscillations in the computed compressive stresses can still be seen with slightly reduced amplitudes, and meanwhile, the wave fronts become less sharp due to increased viscosity dissipation. Recently, the GIMP noise of particle stresses has also been observed in the gas shock problem.[37] The noise of the GIMP and CPDI calculations in[37] and Figure 14 is mainly due to the cell-crossing error, namely, discontinuity or rapid change of the gradient of the shape function gradient in a single time step, which can be characterized by the particle Courant number as defined in.[37] Therefore, in the cases involving large deformation and/or high particle velocity, the GIMP and CPDI improvements in reducing cell-crossing error via the introduction of finite particle domain could be reduced or even lost. The oscillations in compressive stresses in Figure 14 are inherent in the GIMP and CPDI, while the BSMPM outperforms the GIMP and CPDI in terms of the computational accuracy for the impact problems.

We have also solved the problem using the quadratic and quartic BSMPM, in which the parameters are the same as those for the cubic BSMPM computation. Figure 16A compares the stress profiles along the thickness direction at 7.8 ms as obtained with the BSMPM computations using the 3 Gauss point quadrature. The stress waves as computed with the quadratic, cubic, and quartic BSMPM computations are very close. When no Gauss quadrature is invoked, as shown in Figure 16B, little changes in the stress waves predicted by the cubic and quartic BSMPM can be observed, and the wave fronts obtained by all the BSMPM computations are close. Small-amplitude oscillations, however, are visible in the compressive stress wave as computed with the quadratic BSMPM (see the inset). The observed stress oscillations in the quadratic BSMPM obviously arise from the quadrature error, which is considerably reduced as the smoother cubic and quartic B-spline basis functions are used. Therefore, in consistence with the findings in the second example, these

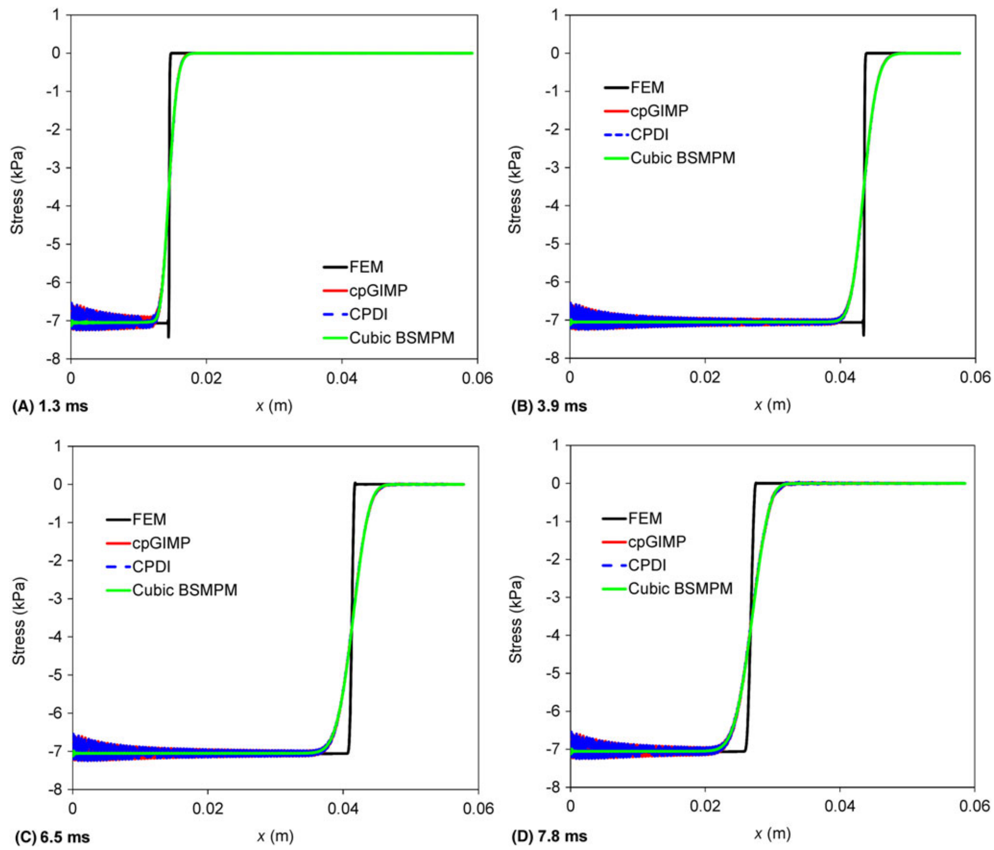

FIGURE 14 Comparisons of the stress profiles along the plate thickness calculated by the cpGIMP, CPDI, and cubic BSMPM at (A) 1.3 ms, (B) 3.9 ms, (C) 6.5 ms, and (D) 7.8 ms.

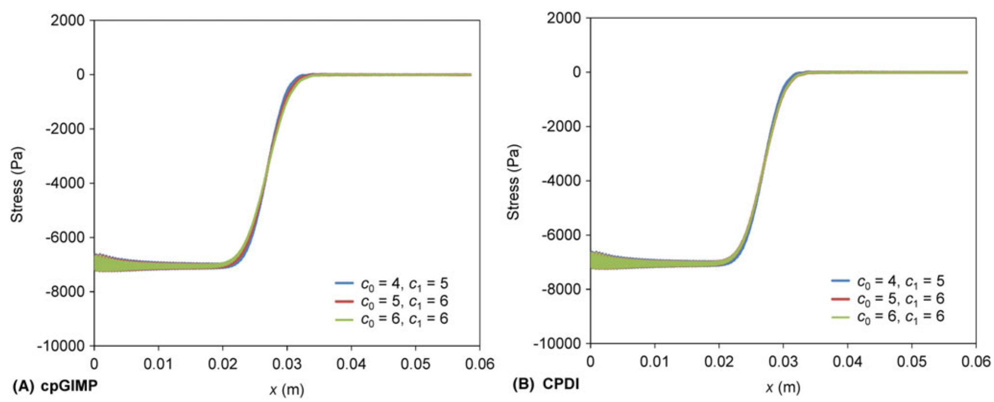

FIGURE 15 The computed stress distributions at 7.8 ms by the cpGIMP and CPDI with different viscosity coefficients of c_0 and c_1

observations also suggest the increase of the B-spline basis function order and/or the adoption of Gauss point quadrature for obtaining the high-accuracy BSMPM solutions to impact problems.

Due to a larger support of B-splines, the BSMPM could yield a more premature detection of multi-field contact than the MPM. Because the only difference between the proposed BSMPM and the MPM is the basis functions, the existing contact algorithm for the MPM is expected to be applicable to the BSMPM. To demonstrate this point, we have simulated the impact of 2 elastic plates using the cubic BSMPM without Gauss quadrature and the cpGIMP, respectively. Figure 17 schematically shows that 2 plates of $H = 0.02$ m in thickness are separated with an initial distance of $l = 0.12$ cm and move towards each other at an initial velocity of $v_0 = 0.6$ m/s. The plates have identical material properties to those in the previous impact example, and all simulation parameters remain unchanged except for $nps = 4$ in all knot spans or grid cells. Likewise, this impact problem is treated as a uniaxial strain case (see Figure 17B). The BSMPM contact

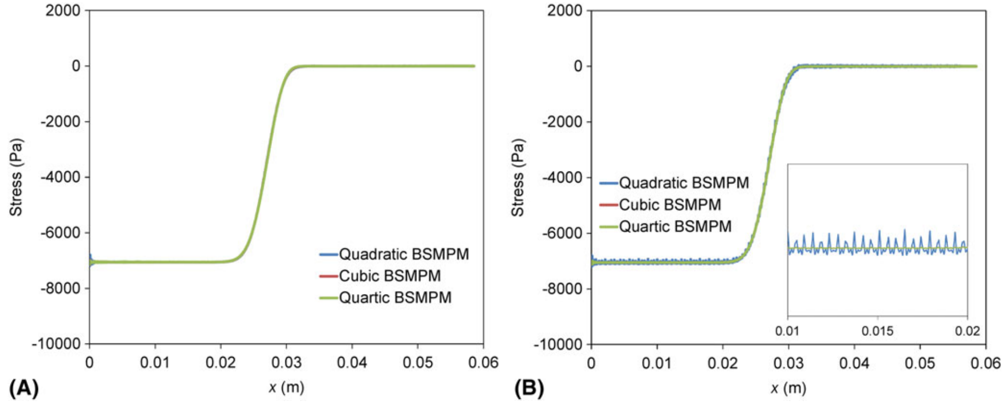

FIGURE 16 Comparisons of stress distributions at 7.8 ms by the quadratic, cubic, and quartic BSMPM computations (A) with and (B) without 3 Gauss point integration. The inset shows the stress distributions in the range of $x = 0.01\text{-}0.02$ m

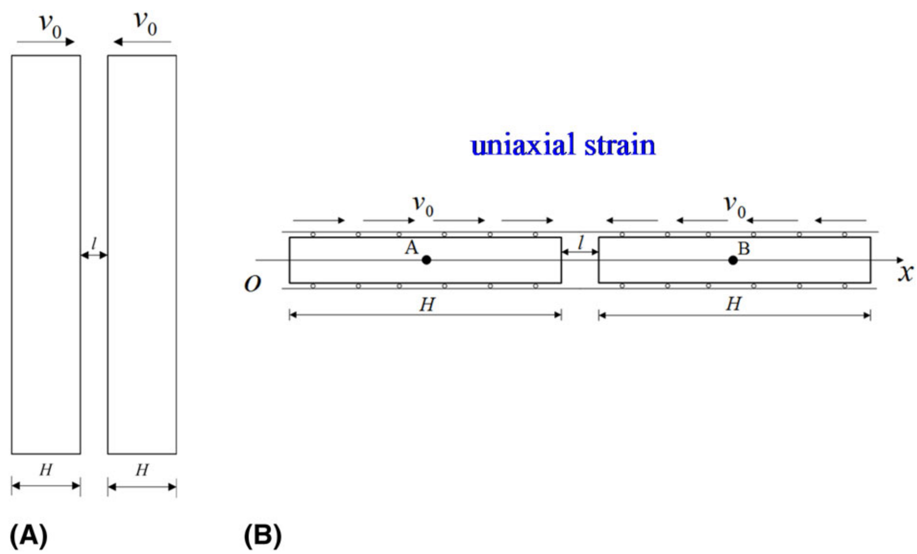

FIGURE 17 Schematic diagram for the impact problem of 2 elastic plates: (A) side view of the problem, and (B) simplified uniaxial strain case along the plate thickness direction

algorithm is implemented in the tensor product grid nodes using the same strategy for the MPM as described in[38-40] to describe the contact and release of the 2 plates. The initial center points A and B in the 2 plates as illustrated in Figure 17 B are monitored, and their transient displacements and velocities by the FEM, cpGIMP, and cubic BSMPM are plotted in Figure 18. It is clearly shown that the cpGIMP and BSMPM results both agree well with the FEM solutions, indicating that the existing MPM contact algorithm can be used for the BSMPM.

4.4 | Spatial convergence of the BSMPM

The dynamic example of a column under self-weight in Subsection 4.2 is considered again in the analysis of the spatial convergence of the BSMPM, for which the knot span sizes are varied and the L^2-error norms for the displacement distribution over the column at 0.5 seconds are calculated using Equation (25). The convergence properties of the BSMPM with and without the Gauss quadrature are both investigated. As the time step is 2.0×10^{-5} seconds for the cases of $\Delta x = 3.125 \times 10^{-3}$ m and larger, smaller time steps of 1.0×10^{-5} and 0.5×10^{-5} seconds are used for $\Delta x = 1.5625 \times 10^{-3}$ and 0.78125×10^{-3} m, respectively. Our numerical tests have illustrated that the differences in the displacements and stresses are less than 0.3×10^{-3} m and 0.8 Pa for the cases of $\Delta x = 1.5625 \times 10^{-3}$ and 0.78125×10^{-3} m, respectively, as their time steps are halved.

The convergence behaviors for the initial discretization of 4 particles per knot span or grid cell are first discussed. Figure 19A presents the spatial convergence of the errors in the displacement for the MPM, cpGIMP, and CPDI

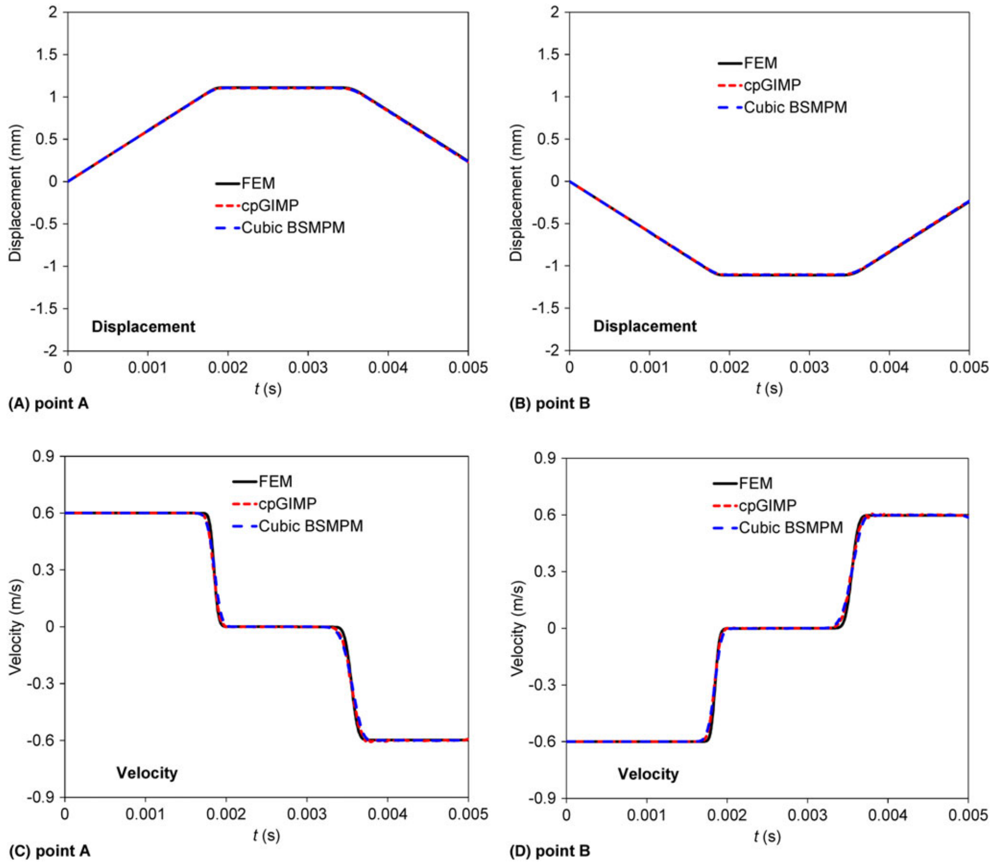

FIGURE 18 Comparisons of the transient displacements and velocities of the initial plate centers obtained with the FEM, the cpGIMP, and cubic BSMPM: (A) displacement of point A, (B) displacement of point B, (C) velocity of point A, and (D) velocity of point B

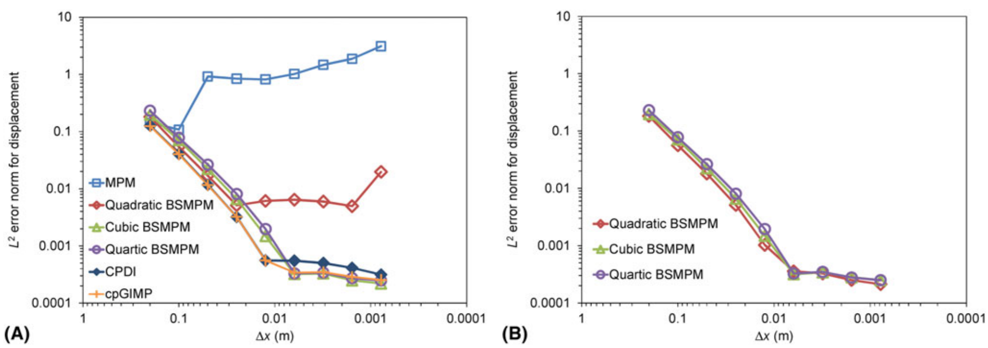

FIGURE 19 Convergence of the L^2 error in the displacement norm for (A) the MPM, cpGIMP, CPDI methods, and the BSMPM without Gauss quadrature, and (B) the BSMPM with Gauss quadrature. The error norms correspond to the particle displacements at time 0.5 s for the problem of a column under self-weight. The MPM computations are performed using the same parameters as those for the BSMPM

approaches, and the BSMPM without the Gauss quadrature. Similar to the earlier studies,[13,14] the original MPM exhibits the convergence for very low grid resolutions corresponding to Δx equal to 0.1 m and larger. The predominant quadrature error accounts for the loss of the MPM convergence at smaller Δx.[14]

As the piece-wise linear basis functions are replaced with smoother B-spline basis functions, the quadrature error is greatly reduced and the BSMPM shows significantly improved convergence. The cubic and quartic BSMPM computations have nearly identical convergence properties, that is, their errors converge at a rate of approximately 2 until $\Delta x = 6.25 \times 10^{-3}$ m and then lies in a plateau stage for Δx as small as 7.8125×10^{-4} m with a significantly reduced convergence rate. For the quadratic BSMPM, a convergence rate close to 2 is also obtained. However, the convergence ceases at a much coarser grid of $\Delta x = 0.025$ m as compared with the results for the cubic and quartic B-splines, and a noticeable rise in the error appears at $\Delta x = 7.8125 \times 10^{-4}$ m, indicating the inferior convergence property of the quadratic BSMPM.

Like the cubic and quartic BSMPM calculations, the cpGIMP and CPDI simulations also present the positive convergence for the grid cell size down to 7.8125×10^{-4} m. At lower grid resolutions corresponding to $\Delta x = 1.25 \times 10^{-2}$ m and

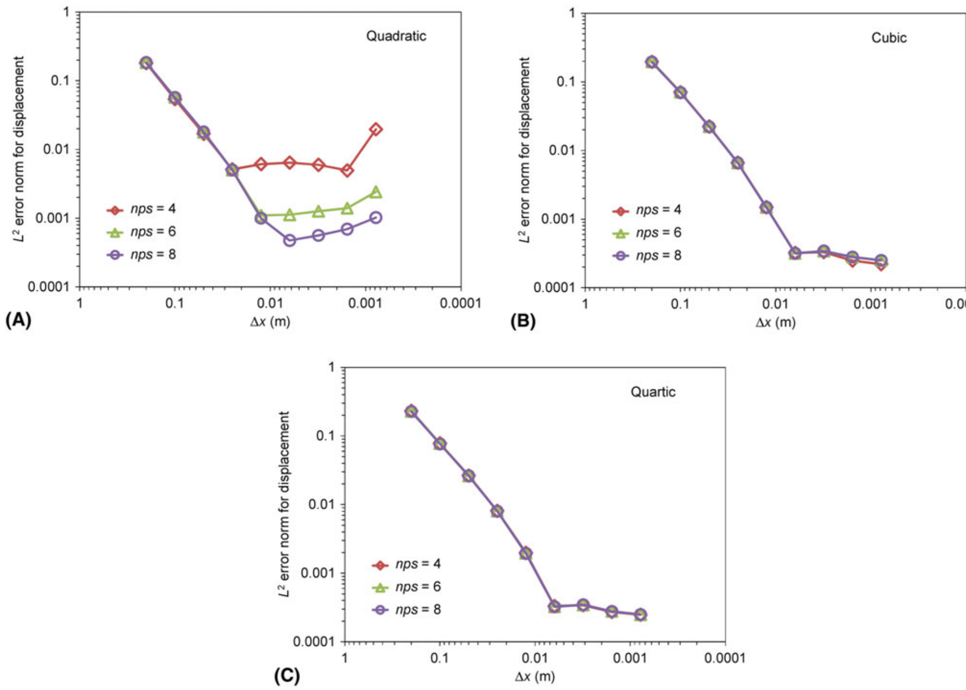

FIGURE 20 Convergence of the L^2 error in the displacement norm of the BSMPM without Gauss quadrature for different initial numbers of particles per knot span: (A) quadratic BSMPM, (B) cubic BSMPM, and (C) quartic BSMPM

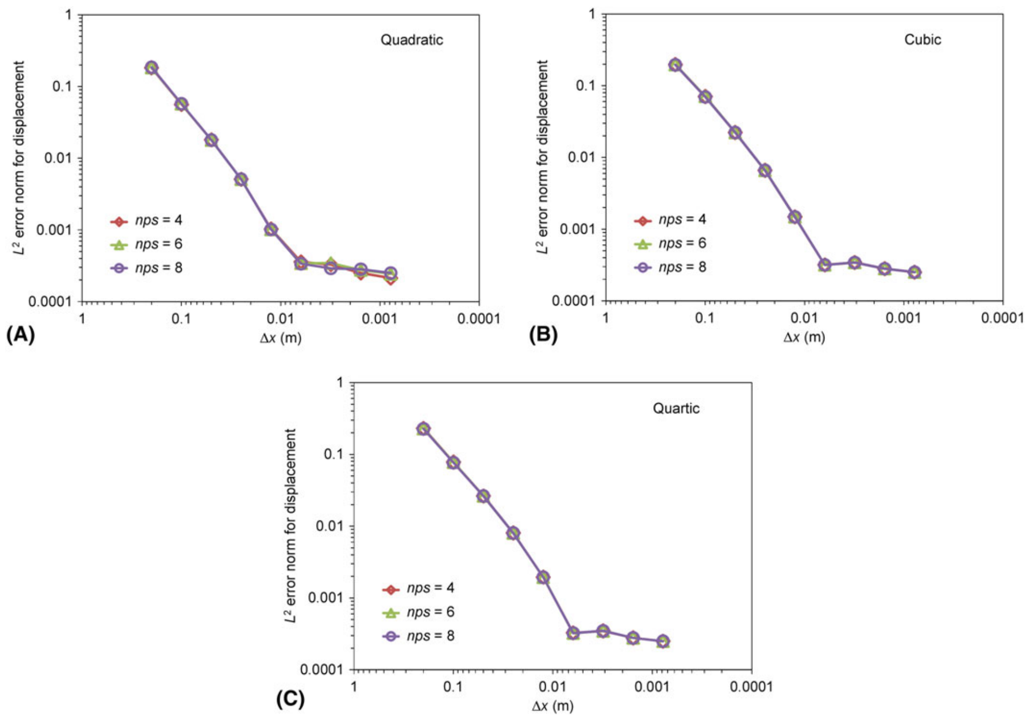

FIGURE 21 Convergence of the L^2 error in the displacement norm of the BSMPM with Gauss quadrature for different initial numbers of particles per knot span: (A) quadratic BSMPM, (B) cubic BSMPM, and (C) quartic BSMPM

larger, a convergence rate of approximately 2 is shown for both cpGIMP and CPDI, and the cpGIMP and CPDI results have a higher accuracy than the BSMPM solutions. As Δx becomes less than 1.25×10^{-2} m, the convergence rates of the CPDI and cpGIMP significantly drop. In addition, at these increased grid resolutions (ie, Δx less than 1.25×10^{-2} m), the CPDI produces lower accuracy results than the cubic and quartic BSMPM, while the cpGIMP results are almost the same as those obtained by the cubic and quartic BSMPM.

When using the Gauss quadrature in the BSMPM, as shown in Figure 19B, the spatial convergence of the quadratic BSMPM is substantially improved, and the almost same convergence behaviors are shown for all the BSMPM computations. Moreover, the comparison between Figure 19A and 19B obviously illustrates no significant improvement on the convergence of the cubic and quartic BSMPM with the use of Gauss quadrature. Therefore, it indicates that the higher-continuity cubic and quartic B-spline basis functions offer higher-accuracy quadrature results, which, in turn, leads to superior convergence properties of the cubic and quartic BSMPM.

The spatial convergences of the BSMPM with increased initial numbers of particles per knot span, namely, $nps = 6$ and 8, are also investigated. Figures 20 and 21 present the corresponding spatial convergences of the L^2 error norm in the displacement for the BSMPM without and with Gauss quadrature, respectively. It is seen that increasing the initial number of particles per span has only a noticeable impact on the convergence of the quadratic BSMPM without the Gauss quadrature. Namely, the convergence is extended to a finer grid of $\Delta x = 1.25 \times 10^{-2}$ and 6.25×10^{-3} m for $nps = 6$ and 8, respectively, versus $\Delta x = 0.025$ m for $nps = 4$. Whereas, the error increases as the span size is further reduced, showing a lack of convergence. In the other cases, insignificant changes in the errors are shown as more particles per span are initialized, and the positive convergence is found for $\Delta x = 7.8125 \times 10^{-4}$ m corresponding to 1280 knot spans. As a result, increasing the initial number of particles per span might not be a feasible way to improve the spatial convergence of the BSMPM.

5 | CONCLUDING REMARKS

The BSMPM that uses quadratic, cubic, and quartic B-spline basis functions within the MPM framework has been developed and evaluated for simulating quasi-static and dynamic problems. With the use of smoother B-spline basis functions, the BSMPM is able to overcome the cell-crossing error resulting from the discontinuous gradient of piece-wise linear basis functions at the grid cell boundary and thus exhibits the higher solution accuracy than the original MPM. The use of B-spline basis functions is found to cause stress oscillations at the boundary. An approach of simply using higher particle density in the boundary span is proposed and demonstrated to be able to greatly diminish such stress oscillations at the boundary. In the small-deformation case, the quadratic, cubic, and quartic BSMPM algorithms all produce the results very close to the analytical solutions and present good energy conservation. For the large-deformation case, however, the quadratic BSMPM scheme yields the stress results in a poor agreement with the FEM and shows the inferior convergence property, as compared with the cubic and quartic BSMPM algorithms. A Gauss quadrature method is proposed with respect to the particle domain, for reducing the quadrature error in the calculation of the internal forces in the BSMPM. It is illustrated that the incorporation of Gauss quadrature could drastically improve the accuracy and convergence of the quadratic BSMPM, while there are no significant differences in the results as obtained by the cubic and quartic BSMPM computations with and without the Gauss quadrature. This finding indicates that the quadrature error is the main mechanism for the lower quality of the quadratic BSMPM for large-deformation problems, and could be effectively reduced by using the B-spline basis functions with higher order and/or the Gauss integration scheme. The similar convergence behaviors have been observed for the cpGIMP, CPDI, and cubic and quartic BSMPM. The plate impact problems have been employed to explore the different features of the BSMPM, GIMP, and CPDI methods. Although all the methods could yield reasonable wave front positions, the BSMPM is shown to outperform the GIMP and CPDI with respect to the accuracy of stress magnitudes. As compared with the cubic and quartic BSMPM methods, the quadratic BSMPM shows the inferior accuracy of stress waves. Furthermore, it is illustrated by the impact between 2 elastic plates that the existing MPM contact algorithm is valid for the BSMPM.

This work has demonstrated that the BSMPM possesses the potential in better solving dynamic problems with large deformations, and is a promising substitute for the MPM, GIMP, and CPDI in simulating impact phenomena. It should be noted that, in addition to the required structured meshes for the construction of B-splines, the use of B-spline basis functions leads to nonlocal communication among grid nodes and increased computational time and storage in the BSMPM. These issues are important in the further development of the BSMPM for promoting SBES. In the present study, the mesh refinement by knot insertion is essentially h-refinement, while the method of using higher-order B-spline basis functions is equivalent to the *k*-refinement as proposed in the isogeometric analysis.[15] A systematic investigation and comparison of the grid refinement strategies with the *h*-, *p*-, and *k*-refinements in the isogeometric analysis are therefore suggested for the future work to enhance the BSMPM.

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